

CS 486/686

Independence and Bayesian Networks II

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Lecture 8

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Learning goals

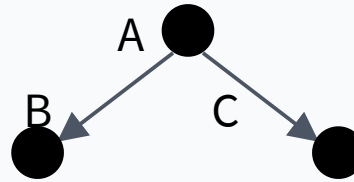
- Apply **d-separation** to check (conditional) independence in any Bayes net
- Given a BN and a variable ordering, **construct a correct Bayes net**
- Distinguish **correlation** from **causation**

Recap: the three structures from L7

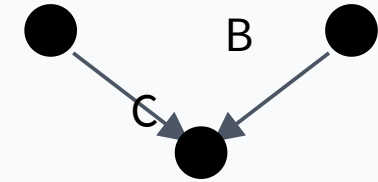
Chain



Common cause



V-structure



Today: generalize the 3 structures to arbitrary paths using d-separation.

D-separation: the rule

D-separation. A set of observed variables E *d-separates* X and Y iff E **blocks every undirected path** between X and Y .

If E d-separates X and Y , then X and Y are **conditionally independent given E** .

A single open path is enough to break independence. We have to check every path and ask: is it blocked?

The three blocking rules

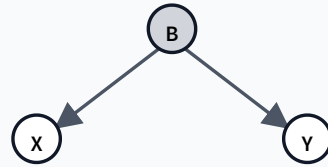
A path through middle node B is blocked iff:

Chain $X \rightarrow B \rightarrow Y$



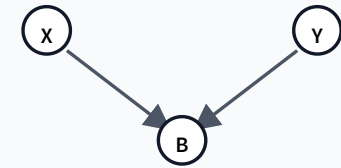
Blocked when $B \in E$
(middle is observed)

Common cause
 $X \leftarrow B \rightarrow Y$



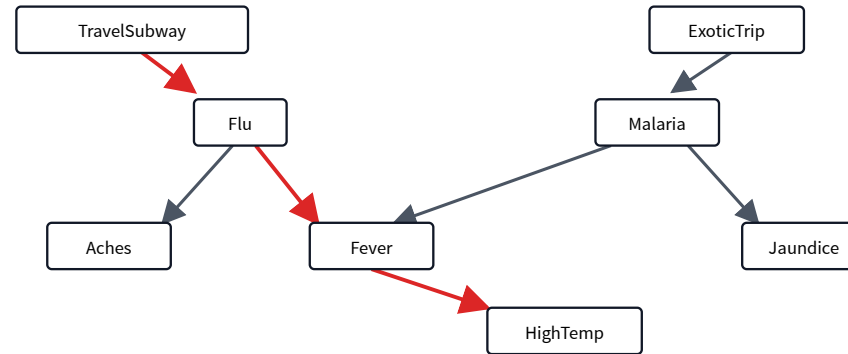
Blocked when $B \in E$
(middle is observed)

V-structure $X \rightarrow B \leftarrow Y$



Reversed: blocked when
neither B nor any of its
descendants are in E

Quiz pack 1: chain blocking



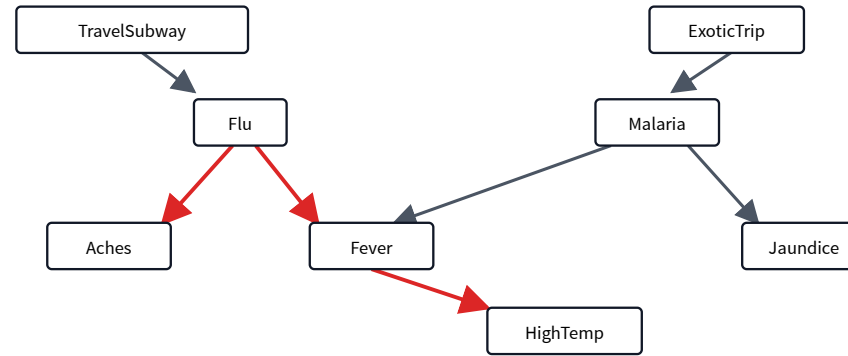
Q1. Are **TravelSubway** and **HighTemp** independent?

No. The path $\text{Subway} \rightarrow \text{Flu} \rightarrow \text{Fever} \rightarrow \text{HighTemp}$ has two chain middles (Flu, Fever); both unobserved $\Rightarrow$ path open.

Q2. Are **TravelSubway** and **HighTemp** independent *given Flu*?

Yes. Observing Flu blocks the chain at the first middle $\Rightarrow$ path closed.

Quiz pack 2: another chain test



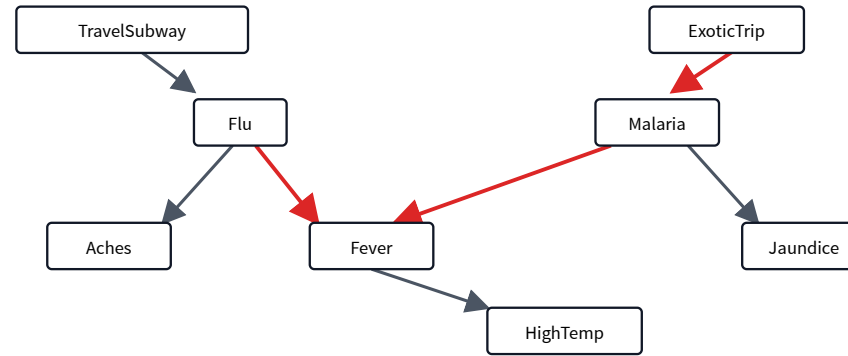
Q3. Are **Aches** and **HighTemp** independent?

No. Path $Aches \leftarrow Flu \rightarrow Fever \rightarrow HighTemp$: common cause (Flu, unobserved) and chain (Fever, unobserved). Both open $\Rightarrow$ path open.

Q4. Are **Aches** and **HighTemp** independent *given Flu*?

Yes. Observing Flu (the common cause) blocks the path at the first middle.

Quiz pack 3: v-structure with a descendant



Q5. Are **Flu** and **ExoticTrip** independent?

Yes. Path $\text{Flu} \rightarrow \text{Fever} \leftarrow \text{Malaria} \leftarrow \text{ExoticTrip}$. Fever is a v-structure middle; neither Fever nor its descendant HighTemp is observed $\Rightarrow$ v-structure blocks the path.

Q6. Independent *given HighTemp*?

No. HighTemp is a descendant of the v-structure middle Fever, so the v-structure now *opens* the path (explaining-away).

Many correct Bayes nets exist

For a fixed joint distribution, more than one Bayes net is "correct".

- **Correctness:** Bayes net B is correct (w.r.t. A) iff every independence that B requires also holds in A .
- **Missing independence is OK** — a denser network still encodes the joint.
- **Missing dependence is NOT OK** — a too-sparse network forces a wrong independence.
- We prefer the BN with the **fewest probabilities** — usually the one with fewest edges.

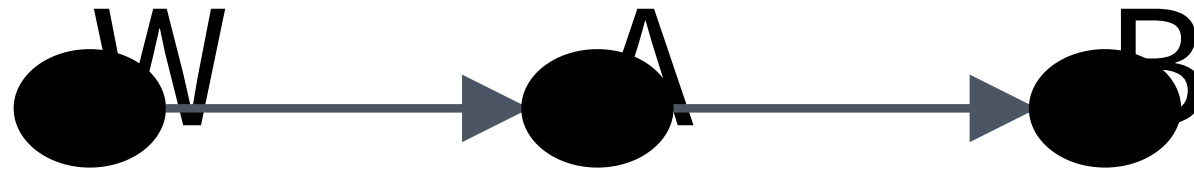
Construction algorithm

1. Order the variables $X_1, \dots, X_n$.
2. For each X_i , choose the **smallest** subset of $\{X_1, \dots, X_{i-1}\}$ such that, given those parents, X_i is independent of the rest.
3. Add edges from each parent to X_i ; write down $P(X_i \mid \text{Parents}(X_i))$.

Ordering matters: a bad order can force every later variable to depend on many earlier ones.

Example 1: chain BN, order W, A, B

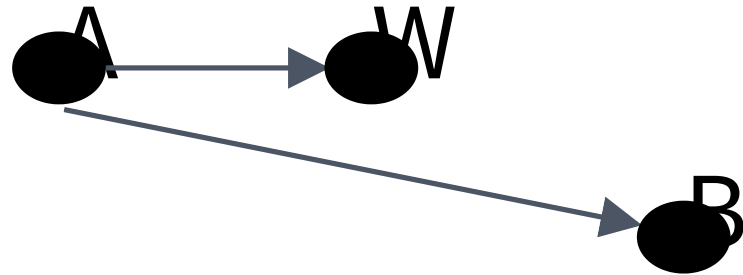
Original BN: 



Final: $W \rightarrow A \rightarrow B$. Same shape as the original (2 edges).

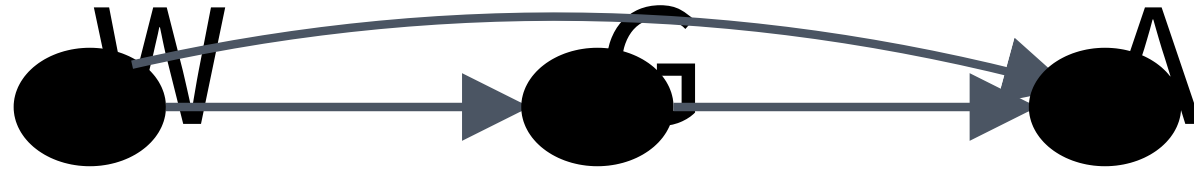
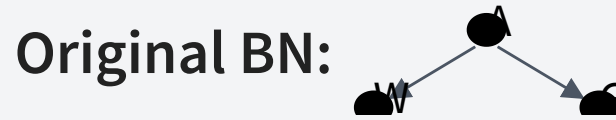
Example 1 alt: same BN, order A, W, B

Original BN: 



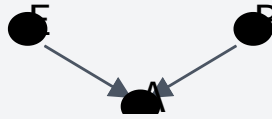
Final: $A \rightarrow W$ and $A \rightarrow B$. Different shape, still 2 edges — *different but equally compact.*

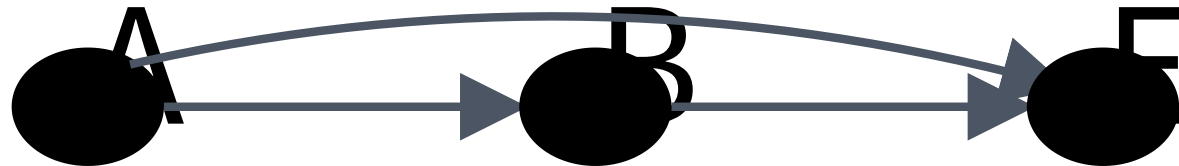
Example 2: common cause, order W, G, A



Final: 3 edges ($W \rightarrow G, W \rightarrow A, G \rightarrow A$) — *more than the original's 2 edges.*
Suboptimal!

Example 3: v-structure, order A, B, E

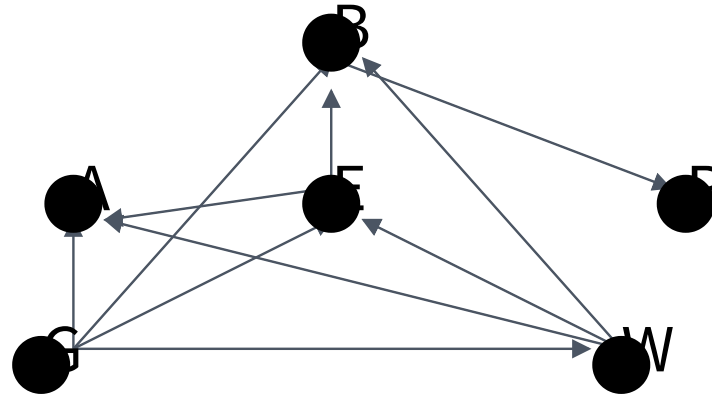
Original BN: 



Final: 3 edges ($A \rightarrow B, A \rightarrow E, B \rightarrow E$) — *more than the original's 2*. Reversed v-structures are expensive.

Holmes with a bad order: G, W, E, B, A, R

Adding effects before causes forces every later node to depend on every earlier one.



$1 + 2 + 4 + 8 + 16 + 2 = 33$ probabilities — vs **12** with the causal order.

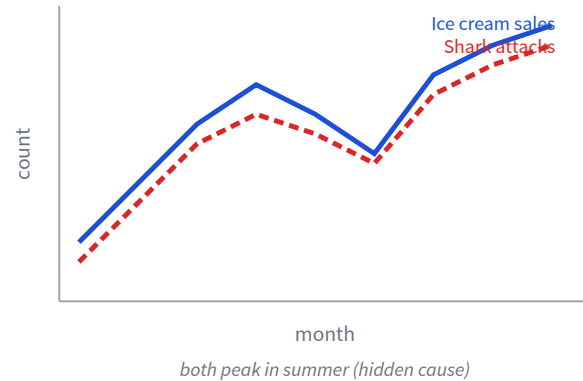
Pick a causal order

Causes precede effects. Add root causes first; effects last.

Example	Original	Order	Reconstructed	#Edges
Ex 1	$B \rightarrow A \rightarrow W$	W, A, B	$W \rightarrow$ $A \rightarrow B$	2 (same)
Ex 1 alt	$B \rightarrow A \rightarrow W$	A, W, B	$A \rightarrow$ $W, A \rightarrow$ B	2 (different)
Ex 2	$A \rightarrow W, A \rightarrow G$	W, G, A	$W \rightarrow$ $G, W \rightarrow$ $A, G \rightarrow A$	3
Ex 3	$F \setminus A \setminus B \setminus A$	$A \setminus B \setminus F$	$A \setminus$	3

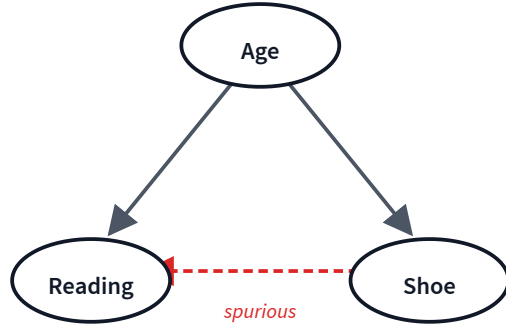
Correlation $\neq$ causation

Two variables can be highly correlated without one causing the other.



- Ice cream sales correlate with shark attacks. Does ice cream attract sharks?
- No — both rise in **summer**. Temperature is a hidden common cause.
- An edge in a Bayes net is associational, *not necessarily causal*.

Confounding variables



- Bigger shoes correlate with better reading scores.
- **Hidden cause:** Age. Older children have bigger feet *and* read better.
- The Shoe → Reading "effect" is **spurious**; controlling for Age would erase it.

Causal intervention: the do operator

Observing vs. *forcing*:

- $P(R|\text{Shoe} = 1)$ — what we see when we *condition* (large feet correlate with reading).
- $P(R|\text{do}(\text{Shoe} = 1))$ — what would happen if we *intervene* and force big shoes onto a random child.

Average Treatment Effect (adjusting for confounders):

$$\text{ATE} = \sum_A P(R|S=1, A) P(A) - \sum_A P(R|S=0, A) P(A) \approx 0$$

$\text{ATE} \approx 0$ tells us shoe size doesn't *cause* reading skill — randomised experiments confirm this.

condition on S $\text{do}(S = 1)$

Learning goals (recap)

- ✓ Apply d-separation to check (conditional) independence
- ✓ Construct a correct Bayes net from a variable ordering
- ✓ Distinguish correlation from causation

Next: decision theory

We can now reason under uncertainty. Next, we combine these probabilities with **utilities** to *act* — choose the action that maximizes expected reward.